

Keyun CHENG

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ABOUT ME

Keyun Cheng received the B.Eng. degree in Software Engineering from Sun Yat-Sen University in 2018, and the M.Sc. degree in Computer Science from The Chinese University of Hong Kong in 2019. He is now a Ph.D. student in Computer Science and Engineering at The Chinese University of Hong Kong. His research interests include distributed storage systems, edge storage systems, multi-cloud storage systems and erasure coding.

EDUCATION BACKGROUND

Aug. 2020 – Current	THE CHINESE UNIVERSITY OF HONG KONG (CUHK) Major: Ph. D. in Computer Science and Engineering	Department of Computer Science and Engineering GPA: N/A	Ranking: N/A
Sep. 2018 – Nov. 2019	THE CHINESE UNIVERSITY OF HONG KONG (CUHK) Major: M.Sc. in Computer Science	Department of Computer Science and Engineering GPA: 3.80/4.00	Ranking: top 3
Sep. 2014 – June. 2018	SUN YET-SEN UNIVERSITY (SYSU) Major: Software Engineering	School of Data & Computer Science GPA: 3.8/4.0	Ranking: 31/122

HONORS AND AWARDS

2020	● CUHK Postgraduate Studentship
2019	● Dean's List Scholarship, M.Sc. in CS, CUHK
2019	● First Prize Winner (top 3) of Distinguished Academic Performance Scholarship, M.Sc. in CS, CUHK
2017	● Second Class Scholarship of SYSU
2017	● Meritorious Winner of Mathematical Contest in Modeling
2016	● First Class Scholarship of SYSU
2016	● Third Prize of China Undergraduate Mathematical Contest in Modeling
2016	● Honorable Mention of Mathematical Contest in Modeling
2015	● Third Class Scholarship of SYSU
2015	● Third Prize of ACM programming competition of SYSU

ACADEMIC EXPERIENCE

Jun. 2019 – Present	Distributed Edge Storage System: Design and Implementation (Supervisor: Prof. Patrick P. C. Lee), CUHK	Group Member
	● We are doing research on distributed storage systems running at edge and clouds. We target to build a storage system to provide (i) low-latency data access from end clients, (ii) fault-tolerance guarantee at both edge and clouds, (iii) data Consistency between edge and clouds, (iv) data security.	
Sep. 2018 – Apr. 2019	Robotic Surgical Instrument Segmentation (Supervisor: Prof. Pheng-Ann Heng), CUHK	Group Member
	● Incorporating temporal information for video segmentation using ConvLSTM and optical flow with standard U-Net-like backbone. Attention map calculated from previous prediction and motion flow may benefit future prediction using a special attention module.	
	● One paper as second author accepted in MICCAI 2019	
Apr. 2019 – May. 2019	Incremental Structure from Motion (SFM) for 3D reconstruction,	Personal Project

CUHK

- Given several images of an object from different viewpoint, incremental SFM starts from reconstructing the 3D model using a carefully selected initial pair of images. Additional images will be fully utilized by triangulation in a reasonable order for better reconstruction results. Bundle adjustment will be performed to minimize re-projection error.

Oct. 2018 – Dec. 2018

Application of SVD for Book Recommendation, CUHK

Project Leader

- Given sparse product review datasets from Amazon, we applied standard Probabilistic Matrix Factorization (PMF), Biased Matrix Factorization (BMF) and SVD++ for book recommendation.
- Implemented using latest Pytorch (0.4.1) in Python.

Mar. 2017 – Jul. 2017

Human Cyber Physical Intelligence Integration Lab (HCPLab) (Supervisor: Liang Lin), SYSU

Member

- Initial research about machine learning, deep learning and High-Performance Computing.
- Designed a basic Parameter Server for distributed machine learning using OpenMP and MPI and deployed it into Caffe.
- Applied the distributed caffe into ‘Tianhe-2’ for testing and achieved significant performance improvement.

Dec. 2016 – Jan. 2017

Design and Development of Android Running App ‘Mop Running’, SYSU

Project Leader

- Developed an Android application that can track users’ speed, distance traveled, and calories burned while even mapping the route.
- Obtained the grade of 96 and ranked the top 6 of the class.

Jun. 2016 – Dec. 2016

Design and Development of Hand-Written Digit Recognition Application, SYSU

Project Leader

- Developed a C++ application using B+ trees and Medrank Algorithm to identify similar hand-written digits. Data will be stored in B+ trees in disk.
- Ranked the top 10 of the class

PUBLICATIONS

- **Fast Predictive Repair in Erasure-Coded Storage: Analysis, Design, and Implementation**
Xiaolu Li, Keyun Cheng, Zhirong Shen, and Patrick P. C. Lee
IEEE Transactions on Parallel and Distributed Systems (TPDS), (under review).
- **Incorporating Temporal Prior from Motion Flow for Instrument Segmentation in Minimally Invasive Surgery Video**
Yueming Jin, Keyun Cheng, Qi Dou, Pheng Ann Heng.
Medical Image Computing and Computer Assisted Intervention (MICCAI), 2019, (Oral)

WORKING EXPERIENCE

Aug 2019 – Jul. 2020

CU Coding Limited

R&D Engineer

- Storage technology R&D, especially the multi-cloud storage solution nCloud and DMS on cloud iNAS.
- Assisting the product maintenance and bug fixes
- Customer project management and support

INTERNSHIP EXPERIENCE

Apr 2017 – Jun. 2017

Research & Development Technology Department of Bigo

- Designed a crash log upload module in Android Client using Google Breakpad.
- Designed a crash analysis system that can be used to collect crash log from Android Client, make analysis. The results of analysis will be showed on web pages.
- Analyzed network abnormality from Android Client.

Dec 2017 – Apr. 2018

AI group of Guangzhou Intelligence Communications Technology Co. Ltd. (Teligen)

- Designed a graphical image annotation tool based on Github

open-source project labelImg. The updated version of labelImg can be used to generate contours for objects in images and auto-label them using up-to-date deep learning technologies. Label files can be stored and transmitted using zeroc-ice.

COMMUNITY ACTIVITIES

Oct. 2015 – June.2018	Network and Information Technology Center of SYSU	Assistant
Sep. 2015 – Sep.2016	School of Data & Computer Science, SYSU	Teaching Assistant
Sep. 2014 – Sep.2016	Music Association of SYSU	Planning Minister
Sep. 2014 – Sep.2015	Student's Software Technology Development Center of SYSU	Secretary
Sep. 2014 – Sep.2015	Student Union of SYSU	Secretary

ADDITIONAL SKILLS

- **English Proficiency:** CET-4: 577 CET-6: 532 IELTS: 7.0 (L: 8.5 R: 7.0 W: 6.0 S: 5.5)
- **Computer Skills:** C++ Python Java
- **Github:** <https://github.com/keyuncheng>
- **Hobbies:** Basketball Long-distance Running Guitar